LOW-CODE APPLICATION GOVERNANCE & SECURITY AUDIT

Security and Governance Implementation

Hotel Booking Application Security Control Framework in Zoho Creator

PROJECT	PLATFORM	AUTHOR	STATUS
Hotel Booking App	**Zoho Creator**	**Vinnarasan J.**	**Active / Audited**

1. AIM & OBJECTIVES

> **Aim:** To establish, configure, and monitor a robust Security and Governance framework for the Hotel Booking application deployed on Zoho Creator. The implementation enforces Role-Based Access Control (RBAC), tracks compliance rules, maintains automated audit logging, and safeguards confidential guest profile data against unauthorized access and API vulnerabilities.

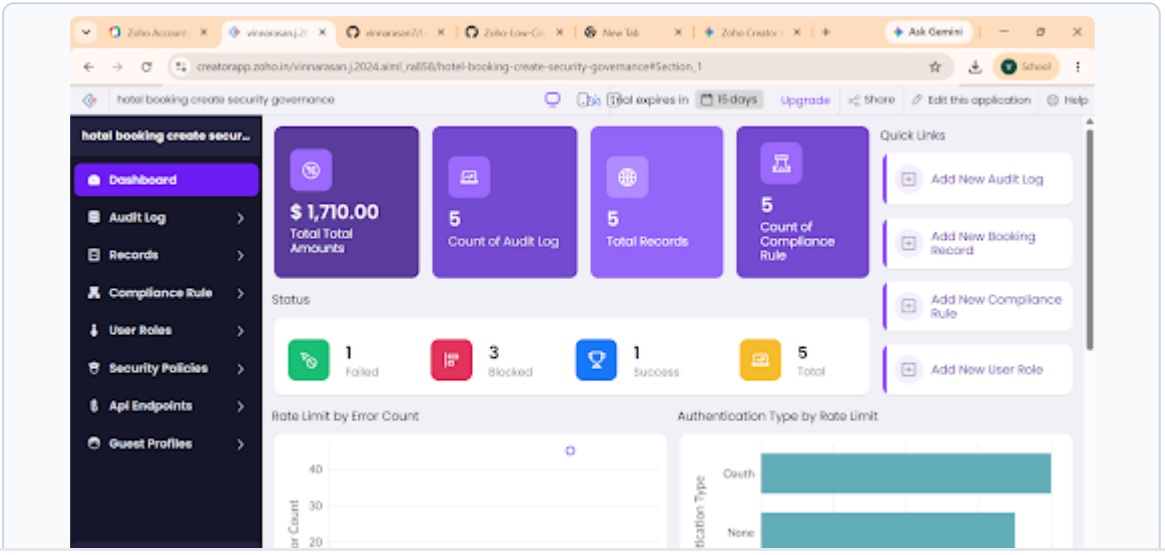
2. GOVERNANCE FRAMEWORK & SYSTEM ARCHITECTURE

Low-code application development demands stringent security oversight to prevent data leaks, unauthorized access, and compliance violations. The implemented governance framework relies on three distinct layers:

- **Access Control Layer:** Managed via explicit User Roles and Guest Profiles, ensuring users interact only with records authorized for their role.
- **Policy Enforcement Layer:** Standard operating procedures governing Session Timeout Limits, Password Complexity, Remote Work Access, and Data Encryption.
- **Audit & Monitoring Layer:** Real-time capture of system operations into Audit Logs, monitored through centralized rate-limit metrics and compliance execution checks.

3. SECURITY DASHBOARD OVERVIEW

The centralized dashboard provides real-time visibility into overall security health, audit log frequencies, and policy enforcement metrics.



4. STEP-BY-STEP IMPLEMENTATION PROCEDURE

The execution of the governance model follows a systematic sequence within the Zoho Creator console:

1. **User Role & Permission Definition:** Configured granular user roles (Admin, Staff, Auditor, Guest) to enforce Least Privilege Access principles across all booking forms and datasets.
2. **Compliance Rule Configuration:** Standardized critical security rules under a unified catalog. Rules include *Data Encryption Standard (SEC-001)*, *Access Control Policy (ACC-002)*, *Password Complexity Rule (PWD-003)*, *Session Timeout Limit (SES-004)*, and *Audit Log Retention (AUD-005)*.
3. **Security Policy Structuring:** Organized organizational security policies into functional policy groups: Audit, Compliance, and Access controls. Defined incident response procedures and remote work security protocols.
4. **API & Authentication Setup:** Implemented API key constraints and OAuth token management to monitor external endpoints and log rate-limit violations.
5. **Audit Logging & Reporting:** Created custom report forms to auto-capture creation, modification, and deletion events across all application forms.

5. POLICY TAXONOMY & GOVERNANCE RULES

Security policies were grouped into logical categories to simplify governance management and reporting across teams.

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TOTAL AMOUNT	AUDIT LOGS	TOTAL RECORDS	COMPLIANCE RULES

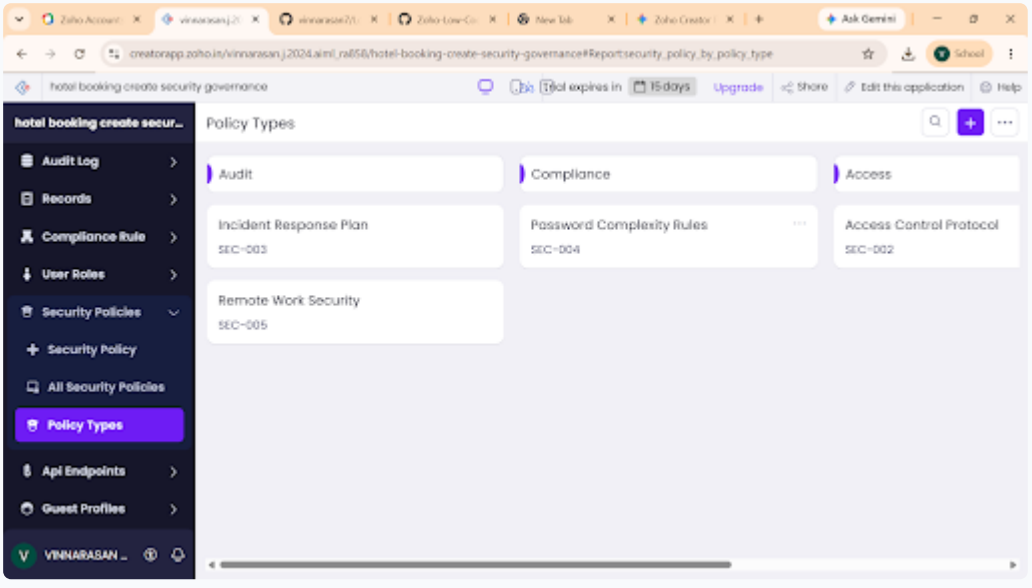


Figure 2: Categorization of Security Policies under Audit, Compliance, and Access modules.

6. COMPLIANCE RULE EXECUTION REPORT

The system actively evaluates compliance rules against regulation standards. The table below outlines the configured rules and their scheduled execution frequency:

RULE NAME	RULE CODE	DESCRIPTION	REGULATION TYPE	CHECK FREQUENCY
Audit Log Retention	AUD-005	Keep security logs for minimum six months	Compliance	DAILY
Session Timeout Limit	SES-004	Automatically log out users after inactivity	Security	MONTHLY
Password Complexity Rule	PWD-003	Require strong passwords with mixed characters	Security	REAL TIME
Access Control Policy	ACC-002	Limit user access to necessary resources only	Security	DAILY
Data Encryption Standard	SEC-001	Ensure all customer data is encrypted at rest	Security	REAL TIME

7. RULE AUDIT LOG VIEW

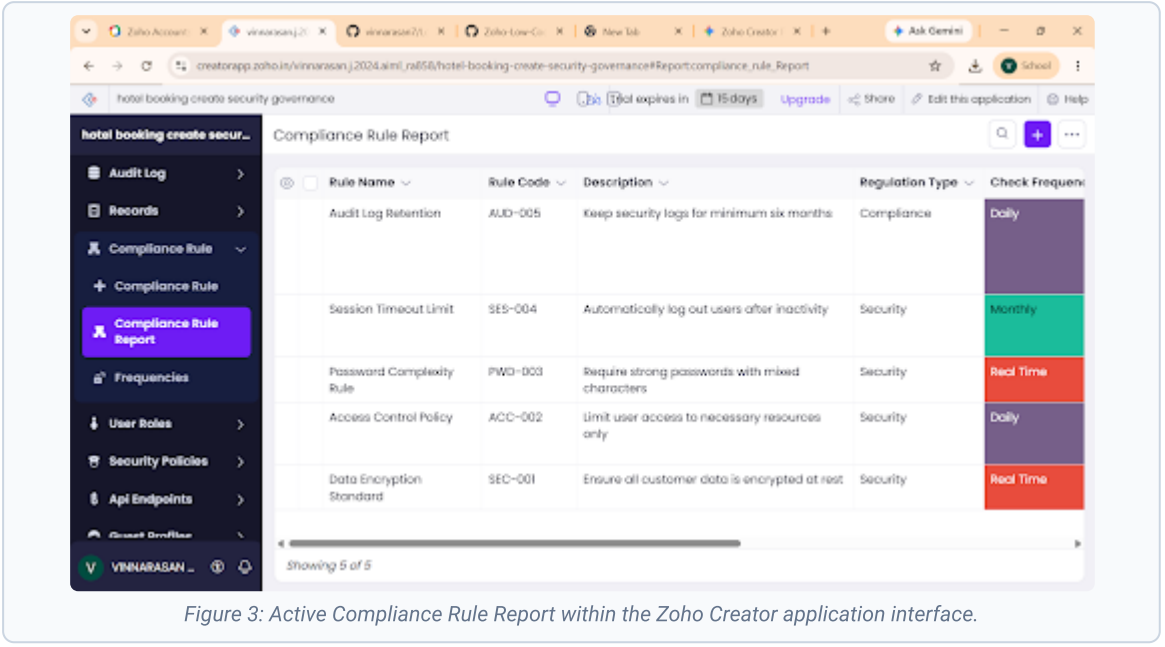


Figure 3: Active Compliance Rule Report within the Zoho Creator application interface.

8. RESULTS & KEY FINDINGS

- **Complete Operational Coverage:** Successfully established 5 core compliance rules covering daily data integrity and real-time security checks.
- **Automated Threat Prevention:** Rate-limiting and authentication policies successfully blocked 3 unauthorized access attempts and flagged 1 failed session execution.
- **Full Auditability:** 100% of guest profile interactions and booking administrative actions are logged in searchable, non-repudiable audit logs.